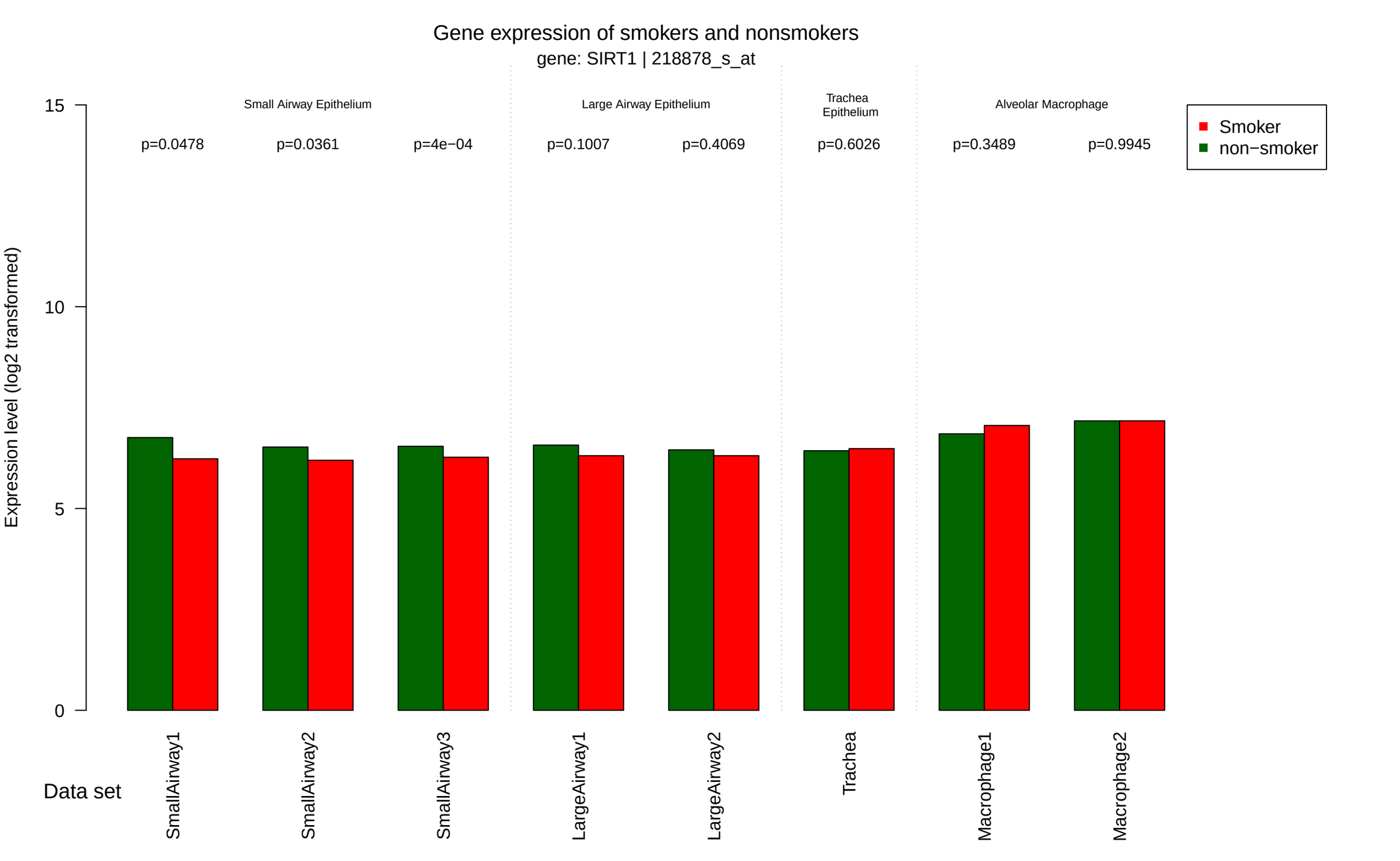
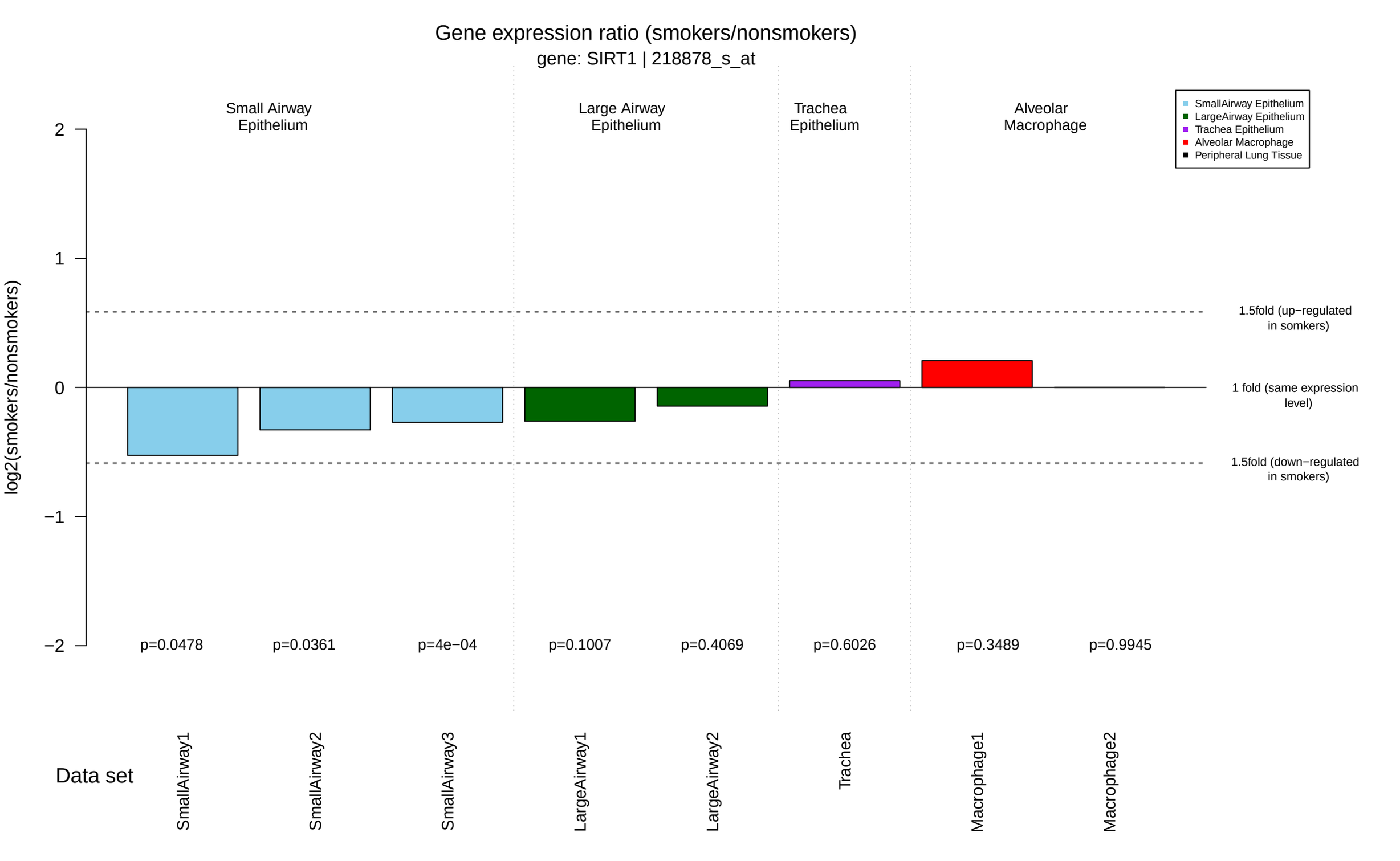
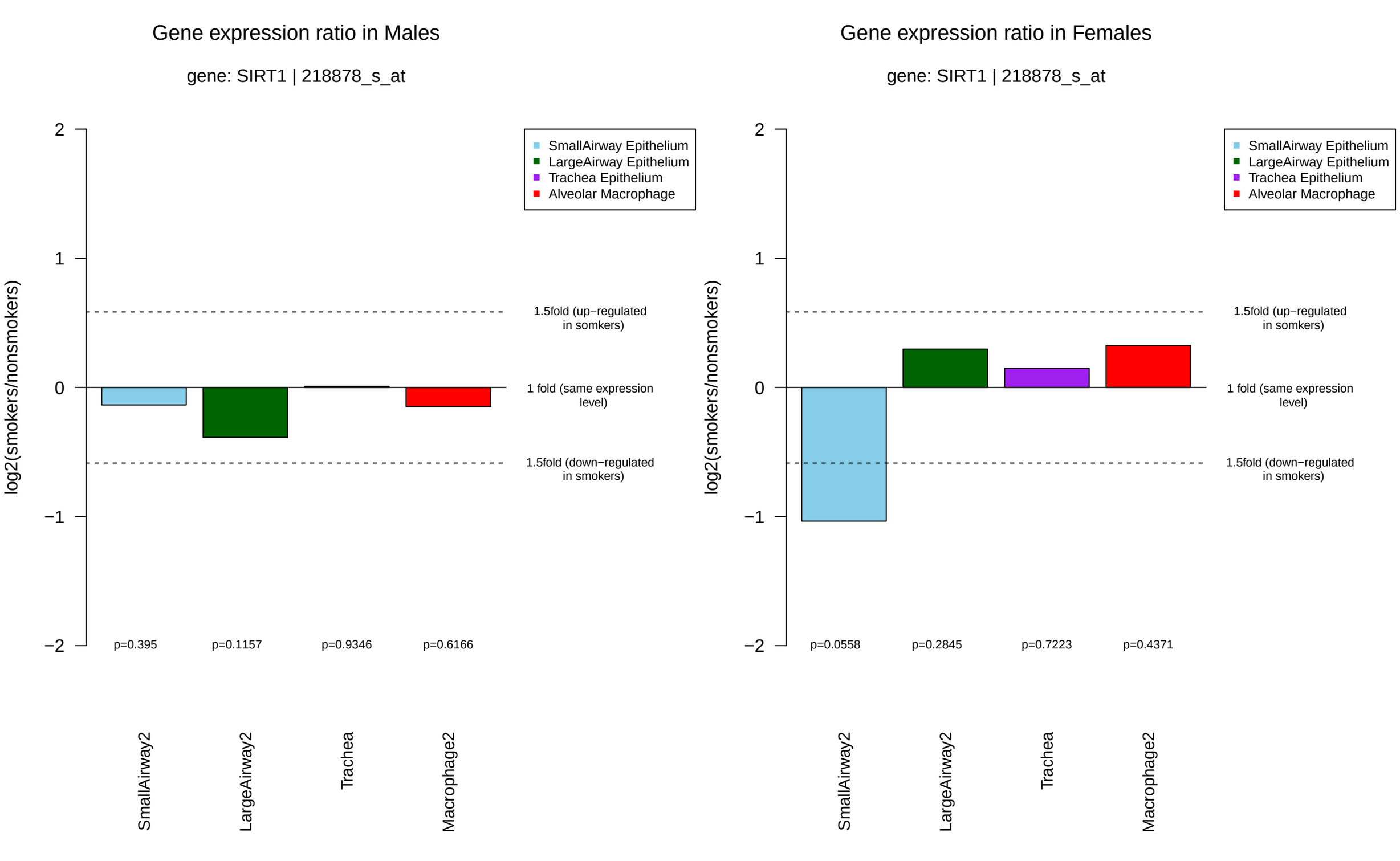
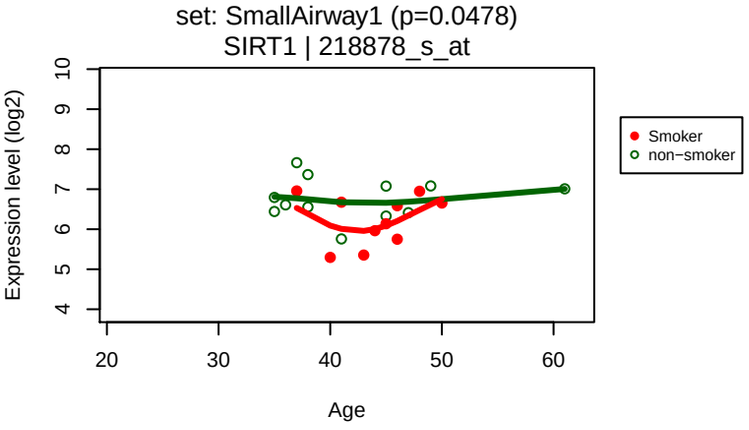
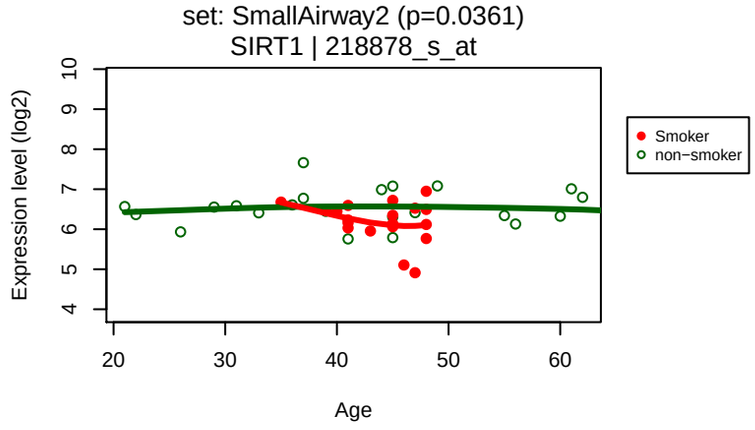
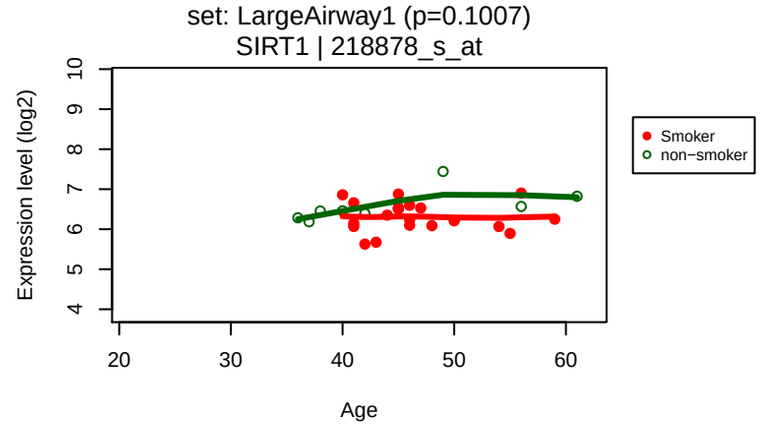

Gene expression of smokers and nonsmokers
gene: SIRT1 | 218878_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0478
p=0.0361
p=4e−04
p=0.1007
p=0.4069
p=0.6026
p=0.3489
p=0.9945
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

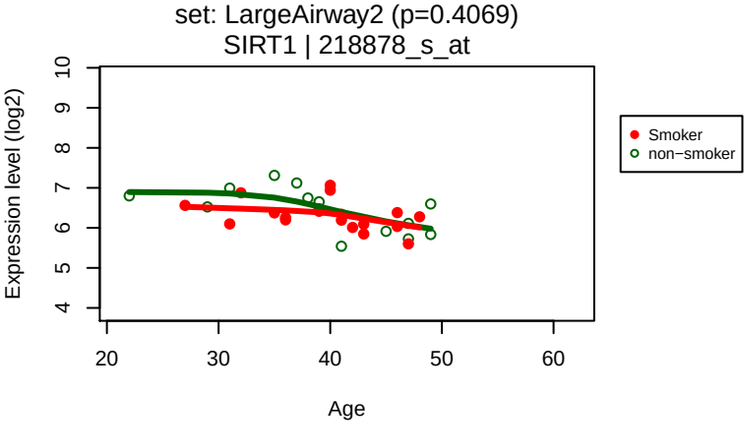

Gene expression ratio (smokers/nonsmokers)
gene: SIRT1 | 218878_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0478
p=0.0361
p=4e−04
p=0.1007
p=0.4069
p=0.6026
p=0.3489
p=0.9945
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

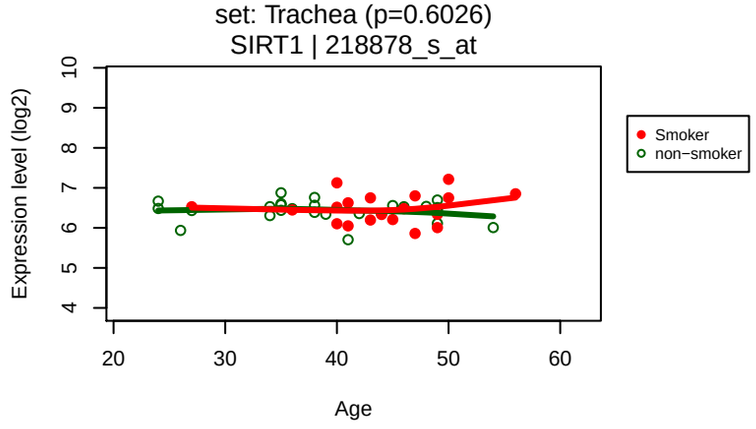

Gene expression ratio in Males
gene: SIRT1 | 218878_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.395
p=0.1157
p=0.9346
p=0.6166
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SIRT1 | 218878_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0558
p=0.2845
p=0.7223
p=0.4371
SmallAirway2
LargeAirway2
Trachea
Macrophage2

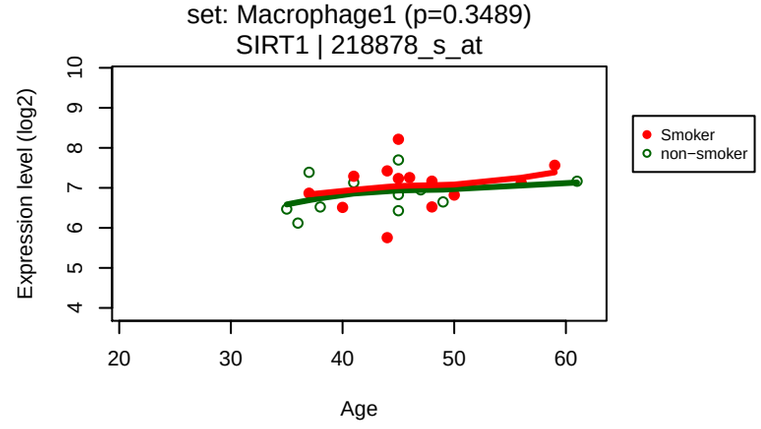

set: SmallAirway1 (p=0.0478)
SIRT1 | 218878_s_at
Expression level (log2)
Age
Smoker
non−smoker

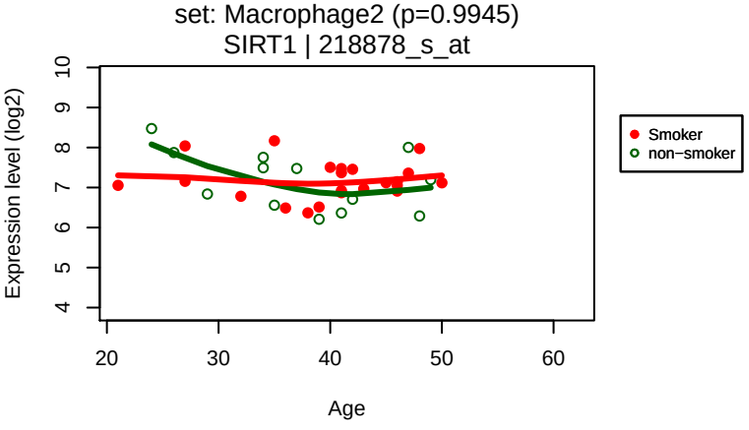

set: SmallAirway2 (p=0.0361)
SIRT1 | 218878_s_at
Expression level (log2)
Age
Smoker
non−smoker

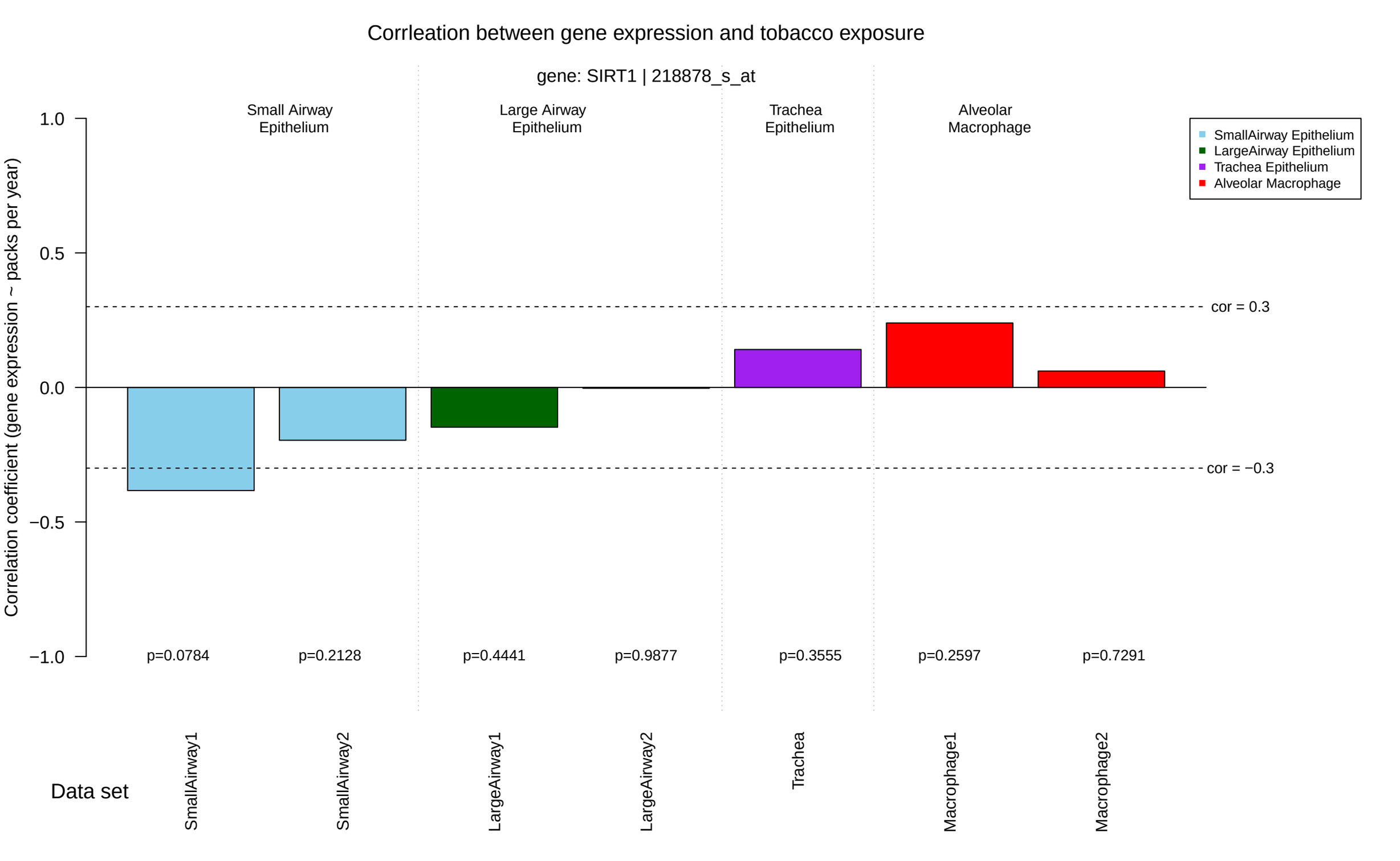

set: LargeAirway1 (p=0.1007)
SIRT1 | 218878_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4069)
SIRT1 | 218878_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6026)
SIRT1 | 218878_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3489)
SIRT1 | 218878_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9945)
SIRT1 | 218878_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SIRT1 | 218878_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0784
p=0.2128
p=0.4441
p=0.9877
p=0.3555
p=0.2597
p=0.7291
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2